

Customer Purchase Behavior Analysis using Python, SQL, and Power BI

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Executive Summary

This project focuses on analyzing customer purchasing behavior using Python, SQL, and Power BI to derive valuable business insights. The analysis aimed to identify key trends influencing customer purchases, evaluate the impact of discounts, and understand subscription patterns. The integration of SQL for data analysis, Python for preprocessing, and Power BI for visualization enabled a complete data-driven approach to improving sales, engagement, and decision-making.

Business Problem

A leading retail company seeks to understand customer shopping behavior to improve revenue, customer satisfaction, and retention. The management observed fluctuating purchase patterns across demographics, products, and sales channels. The objective is to uncover the factors such as discounts, ratings, and payment preferences that influence customer decisions and repeat purchases. The ultimate goal is to leverage data insights to enhance marketing strategies and optimize product and pricing decisions.

Methodology

The analysis was performed in three key stages integrating Python, SQL, and Power BI for end-to-end analytics.

1. Python – Data Preparation and Cleaning

- Loaded and explored a dataset of 3,900 transactions.
- Cleaned missing data in review ratings by imputing median values per category.
- Created new derived columns such as `age_group` and `purchase_frequency_days`.
- Renamed columns for consistency and removed redundant ones.
- Connected Python to PostgreSQL using SQLAlchemy and Psycopg2 to load cleaned data.

2. SQL – Data Analysis and Insights

- Ran structured SQL queries on the `customer_behavior` table to derive insights.
- Analyzed revenue by gender, subscription status, and age group.
- Identified top-performing products and discount effectiveness.
- Segmented customers based on purchase frequency and loyalty.
- Used window functions and aggregation for comparative and categorical insights.

3. Power BI – Visualization and Dashboard

- Developed an interactive Power BI dashboard to visualize key business metrics.
- Implemented slicers for gender, category, and shipping type to allow dynamic filtering.
- Showed KPIs such as customer count, total revenue, and average purchase amount.
- Enabled the business to monitor high-revenue segments, product performance, and shipping trends in real time.

Tools Used

- Python (Pandas, NumPy, SQLAlchemy, Psycopg2)
- PostgreSQL (SQL Querying and Data Management)
- Power BI (Dashboard and Visualization)
- Google Colab / Jupyter Notebook (Scripting and Analysis)
- GitHub (Version Control and Project Repository)

Key Findings and Insights

- Gender-based revenue analysis revealed significant spending differences.
- Customers using discounts often had above-average purchase amounts, proving promotional success.
- Subscribed customers contributed higher average spending and revenue than non-subscribers.
- Express shipping correlated with higher spending, indicating premium user preference.
- Top-rated and frequently purchased products provided a clear picture of customer preferences.
- Returning and loyal customers formed the majority, showing strong brand trust.
- Middle-aged and adult age groups contributed most to total sales revenue.

Business Recommendations

1. Focus marketing campaigns on high-value segments identified in Power BI.
2. Encourage subscriptions by offering loyalty rewards and exclusive discounts.
3. Maintain sufficient stock for top-performing products to prevent lost sales.
4. Adjust discount strategies to improve conversion while maintaining profit margins.
5. Continuously monitor the Power BI dashboard to refine business decisions.

Challenges and Learnings

- Faced PostgreSQL connection issues while integrating Python and SQLAlchemy.
- Managed data inconsistencies and null values during cleaning.
- Debugged SQL syntax and type conversion errors.
- Learned to merge insights from multiple tools for cohesive reporting.
- Improved understanding of end-to-end analytics workflows from data collection to visualization.

Future Scope

- Implement predictive models using machine learning to forecast customer churn.
- Add real-time data integration to Power BI for live business monitoring.
- Expand the dataset to include customer feedback and marketing response data.
- Automate periodic SQL reporting using ETL pipelines for ongoing insights.

Contact and Project Links

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